

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

PivotTable Recommended Table Forms Pictures Shapes Icons SmartArt 3D Models Checkboxes Recommended Charts Maps PivotChart Line Column W/ Loss Sparklines Slicer Timeline Links Comment Text Box Header Wordart Sig

Tables Illustrations Controls Charts

1. eite any examples in grey italic text that are not relevant/applicable to the study once the programming specification has been finalized

Section A: Inclusion / Exclusion Criteria
Note to Programmer: Use the naming convention 'TimePeriod_Variable'

Variable	Label	Values	Definition	Code Lists Group	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
INDEX_DATE	Index Date	date	date of the first medical claim for multiple myeloma.	Tab 42	At least one inpatient medical claim with a diagnosis code for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) OR ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) in any position on claim, on separate days within 30/60/90 days. (Suggest to program this as a macro, where 30, 60 & 90 day windows can be applied to the outpatient rule). The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in tab		OK19012026
INDEX_YR	Index Year	date	year(index date)	Tab 42			OK19012026
AGE_DX	Age year of index numeric	numeric	year(index date) - birthyear	N/A			OK19012026
CE_b	Baseline Continuous Enrollment	Binary	If six months of CE with medical and pharmacy benefits before the index date then =1; else =0;	N/A	Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.		OK19012026
CE_f	Follow-up Continuous Enrollment of at least 1 day	Binary	Only 1 day of CE is required in the follow up period then =1; else =0;	N/A	Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled		OK19012026
CE_3mosf	Follow-up Continuous Enrollment of at least 3mos	Binary	Please flag patients if ≥ 3 months of CE starting on the index date and ending on the earliest of death, health plan disenrollment, or the study period end for patients who do not have an assigned date of death < 3 months after the index date then =1; else =0;	N/A	For patients with an assigned date of death < 3 months after the index date, the follow-up period ended at the assigned date of death. No gaps in enrollment are allowable. This patients may be subset on in the future. Patients should have no gaps in enrollment for continuously enrolled days that exceed 3mos.		OK19012026
MM_FU_agents	Evidence of treatment with agents for the purpose of treating MM	Binary	Evidence of an FDA-approved MM oncology therapy in claims during the follow-up period =1; else =0; Note : Receipt of corticosteroids only (eg dexamethasone) will not be considered evidence of receiving an approved MM oncology therapy	Tab 41	These will be identified with pharmacy claims (NDC) and medical claims (CPTs and HCPCS) codes.		OK19012026
MM_dl_agents	oncology therapy in claims during the baseline period	Binary	≥ 1 medical or pharmacy claim for FDA-approved MM oncology therapy during the baseline period =1; else =0;	Tab 41			OK19012026

File Home Insert Draw Page Layout Formulas Data Review View Automate Help
 PivotTable Recommended PivotTables
 Tables
 Illustrations
 Controls
 Recommended Charts
 Charts
 Maps
 PivotChart
 Sparklines
 Filters
 Links
 Comments
 Text Box & Footer
 Wordart

	B	C	D	E	F	G	H	I
14	MM_baseline_diag	✚ Evidence of any MM during the baseline period	Binary	Evidence of any MM (each of newly diagnosed and R/R MM), during the baseline period =1; else =0.	Tab 42 & 49	Evidence of any MM here is defined use ≥1 non-diagnostic medical claim for MM (ICD-9-CM=203.0x) or ICD-10-CM code=C90.0x) during the baseline period. This flag is used to help better identify smoldering patients. From Optum: "When we refer to a non-diagnostic multiple myeloma (or any disease) claim, what we are referring to is a claim where one of the multiple myeloma diagnosis codes is present, but there is at least one service line on the claim that doesn't equal a diagnostic code. The idea is that a physician/facility might order labs for testing/diagnostics for multiple myeloma and on the claim they have to put something for the diagnosis so they will put what they ordered the lab to test for. e.g., diagnosis code for multiple myeloma. If the resulting facility claim had a diagnosis code for multiple myeloma and only one service line on the claim and that service line indicated a laboratory test, then we can't be certain that this is a true multiple myeloma diagnosis because the claim is suggesting that only testing was done. Alternatively, if the claim had the diagnosis codes for multiple myeloma and two service lines, the first line for testing and the second line a HCPCS code for medication or a CPT code for physician care management then we will accept that claim as a "non-diagnostic claim" indicating multiple myeloma." Patients with either ≥ 1 inpatient or ≥ 2 outpatient ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer. Only the first of the 2 codes is required to occur inside the baseline period. Also flag patients with ≥ 2 outpatient claims on separate days within 60 or 90 days for consideration		OK19012026
15	MM_baseline_other	Evidence of another cancer in the baseline period	Binary	Evidence of another cancer in the baseline period =1; else =0.	Tab 44			OK19012026
16	MM_FU_other	Evidence of another cancer in the FU period	Binary	Evidence of another cancer in the follow up period =1; else =0.	Tab 44	Patients with ≥ 2 ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer.		
17	Pregnant	Evidence of pregnancy	Binary	Evidence of pregnancy =1; else =0.	Tab 45	≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the patient identification period. (baseline or follow-up)		OK19012026

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelp

Pivotal Recommended Pivotal Tables

Pictures Shapes Icons SmartArt

Illustrations

Controls

Recommended Charts

Charts

Maps PivotChart

Line Column W/ Loss

Slicer Timeline

Link

Comment

Text Header Box & Footer

Tables

3D Models

SmartArt

Checkboxes

Recommended Charts

Charts

Sparklines

Filters

Links

Comments

Text

B		C		D		E		F		G		H		I	
CT_part	Clinical trial participation	Binary	Clinical trial participation =1, else =0;	Tab 46	Evidence of clinical trial participation during the patient identification (study period) will be flagged.										
Death_dt	Death Date ✚	Date	Date patient died.		When date of death is only available at month-level granularity, the date will be generalized to the middle of the month or year, respectively (e.g., a date of death in June 2014 will be treated as 15 June 2014). For patients with only death year available, the death date is assumed to be July 15th year of the year they died. For patients with only death year available and their year of death is the same year as their index date, the death date is assumed to be July 15th of that year if the index date is on or before July 15th. The death date is assumed to be December 31st of that year if the index date is after July 15th. Note: When the flag CE_3mosf=1 is applied health plan disenrollment or the study period for patients who do not have have an assigned date of death <3mos after the index date.										OK19012026
ENDDATE	Follow Up End Date	Date	min(Death_dt, study period end date)		When the flag CE_3mosf=1 is applied, follow up end date will be earliest of death date, health plan disenrollment or the study period for patients who do not have have an assigned date of death <3mos after the index date.										OK19012026
ENDDATE_CE	Follow Up End Date among CE pts	Date	min(Death_dt, date of health plan disenrollment, study period end date)		When the flag CE_3mosf=1 is applied, follow up end date will be earliest of death date, health plan disenrollment or the study period for patients who do not have have an assigned date of death <3mos after the index date.										OK19012026
FU_DAYS	Length of follow up period (days)	numeric	The number of days between a patient's ENDDATE and their INDEX DATE + 1												OK19012026
FU_DAYS_CE	Length of follow up period (days) among CE pts	numeric	The number of days between a patient's ENDDATE_CE and their INDEX_DATE + 1		Follow up time among patients who are continuously enrolled (without gaps in enrollment) in the FU period.										OK19012026

Section B: IE Criteria

Note to Programmer: Apply each criteria as a flag that can be carried forward or merged onto a later dataset. Helps make the cohort dynamic and flexible to change future study requirements.
ly criteria in the order specified by the criteria number. Ensure that all flag are able to implemented independently rather than be executed sequentially/interdependent

Criteria Type	Criteria Description		Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
INCLUSION	≥ 1 medical claims for multiple myeloma in any position on claim		Serves as the base of the entire cohort. This flag will always be applied. In the absence of criteria 1 being applied, the earliest medical claim for multiple myeloma will serve as the index date.		OK19012026
INCLUSION	At least one inpatient medical claim with a diagnosis code for multiple myeloma (any ICD-9-CM=203 0x or ICD-10-CM code=C90 0x) OR ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203 x or ICD-10-CM code=C90 x) in any position on claim, on separate days within 30/60/90 days		If criteria 1 flag is not implemented, future IE criteria will rely on the date of a patient's earliest MM claim in the Optum database as the index date.		OK19012026
INCLUSION	Patients at least 18 years of age in the index year				OK19012026
INCLUSION	Patients are required to have evidence of an FDA-approved MM oncology therapy in claims		Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled		OK19012026
INCLUSION	Turn on the following period See tab 41 MMA Criteria				OK19012026
INCLUSION	1.Cover 2.QC Review 3.Data Prep 4.StudyPop 5A.MMA_MED 5B.MAP_MED 6.LOT1_BASE 7.SCT 10.LOT1_BASE_END OK 40.CL MMA ROLLUP 41.CL MMA CODELIST 41a.MMA NDC				OK19012026

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

PivotTable Recommended PivotTables

Tables

Pictures Shapes Icons SmartArt

Illustrations

3D Models

Checkboxes

Recommended Charts

Charts

Maps

PivotChart

Sparklines

Line Column Win/Loss

Slicer Timeline

Filters

Link

Comment

Text Box

Header & Footer

Text

L38		B		C		D		E		F		G		H		I	
33	INCLUSION					Patients are required to have evidence of an FDA-approved MM oncology therapy in claims during the follow-up period. See tab CL MMA CodeList.				Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled						OK19012026	
34	EXCLUSION					≥ 1 medical or pharmacy claim for an FDA-approved MM oncology therapy during the baseline period (9mos prior to index date). See tab CL MMA CodeList.				Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled						OK19012026	
35	INCLUSION					Flag patients with ≥ 6 months of Continuous Enrollment with medical and pharmacy benefits before the index date. The baseline period is fixed and is the six months before the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.										OK19012026	
36	INCLUSION					Flag patients with ≥ 1 day of Continuous Enrollment starting on the index date. Follow up period ends on the earliest of death or the study period end.				Patients with ≥ 3 months of CE from the index date (with no allowable gaps in enrollment) will be flagged for consideration for sensitivity or secondary analyses, and ending on the earliest of health plan disenrollment or the study period for patients who do not have an assigned date of death < 3mos after the index date. For patients with an assigned date of death < 3 months after the index date, these patients are retained and have a limited follow-up period that ends at the assigned date of death.						OK19012026	
38	EXCLUSION					Patients with ≥ 2 non-diagnostic diagnosis codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer				This is evidence of another cancer in the baseline period. Only the first of the 2 codes is required to occur inside the baseline period.						OK19012026	
39	EXCLUSION					≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the baseline or follow-up period. See tab CL PREG.										OK19012026	
40	EXCLUSION					Evidence of clinical trial participation during each of the baseline and follow-up periods. See tab CL CLINTRAL.										OK19012026	

Section C: Final Study Population

Label	Definition	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
43	The team will begin with the requirement that patients have at least one inpatient medical claim with a diagnosis code for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) OR ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.x or ICD-10-CM code=C90.x) in any position on claim, on separate days within 90 days, during identification period. The team will apply all other IE criteria with the exception of the following exclusion criteria: 1) Evidence of another cancer (criteria 7) 2) Pregnancy or childbirth during the baseline or follow-up period (criteria 8) 3) Evidence of RCT participation during the baseline and/or follow-up periods (criteria 9) *Patients that meet exclusion criteria 1-3 indicated here will be included in this build of the algorithm.	Apply each criteria as a flag that can be carried forward or merged onto a later dataset. Helps make the cohort dynamic and flexible to change future study requirements. Suggest to establish global variables upfront when creating inputs for the study population, or other approaches to make the initial inputs easy to change. Suggest to implement IE criteria using a function where follow up period, allowable gaps, etc. can be modified. There is only 1 day of CE required in the follow up period for this initial cohort build. Patients with ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) in any position on claim, on separate days within 30 and 60 days during identification period will be flagged in the attrition table as well. ELIG. COH dataset should include patient ID, index date, death date, study end date, number of follow up days, and all IE criteria. These should be retained		OK19012026

A patient level dataset containing those patients for whom we would like to run the multiple myeloma LOT algorithm on. It is used as an input into the multiple myeloma LOT algorithm.

L49		B	C	D	E	F	G	H	I
A patient level dataset containing those patients for whom we would like to run the multiple myeloma LOT algorithm on. It is used as an input into the multiple myeloma LOT algorithm.		The team will begin with the requirement that patients have at least one inpatient medical claim with a diagnosis code for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) OR ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.x or ICD-10-CM code=C90.x) in any position on claim, on separate days within 90 days, during identification period.		The team will apply all other IE criteria with the exception of the following exclusion criteria: 1) Evidence of another cancer (criteria 7) 2) Pregnancy or childbirth during the baseline or follow-up period (criteria 8) 3) Evidence of RCT participation during the baseline and/or follow-up periods (criteria 9) *Patients that meet exclusion criteria 1-3 indicated here will be included in this build of the algorithm.		Apply each criteria as a flag that can be carried forward or merged onto a later dataset. Helps make the cohort dynamic and flexible to change future study requirements. Suggest to establish global variables upfront when creating inputs for the study population, or other approaches to make the initial inputs easy to change. Suggest to implement IE criteria using a function where follow up period, allowable gaps, etc. can be modified. There is only 1 day of CE required in the follow up period for this initial cohort build. Patients with ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) in any position on claim, on separate days within 30 and 60 days during identification period will be flagged in the attention table as well. ELG, COH dataset should include patient ID, index date, death date, study end date, number of follow up days, and all IE criteria flags should be retained.		OK19012026	